PHYS 6031

Chapter 4: Electrostatics

Multipoles and Macroscopic Media

Spring 2024

OHIO UNIVERSITY

Department of physics + astronomy OHIO UNIVERSITY

Multipole expansion

- Assume that we are "outside" of a localized charge distribution and that $\Phi(r \to \infty, \theta, \phi) \to 0$. Then we expect

$$\Phi(\vec{\boldsymbol{x}}) = \frac{1}{4\pi\epsilon_0} \sum_{\ell m} \frac{4\pi}{2\ell+1} q_{\ell m} \frac{Y_{\ell m}(\theta, \phi)}{r^{\ell+1}}.$$

  Note that ρ need only vanish "fast enough" as $r \to \infty$, e.g., exponentially.

- Recall

$$\Phi(\vec{\boldsymbol{x}}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{\boldsymbol{x}}\,')}{|\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,'|}\, d^3x'.$$

- Using the spherical harmonic expansion for $1/|\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,'|$, we find

$$q_{\ell m} = \int_V \rho(\vec{\boldsymbol{x}})\, r^\ell\, Y^*_{\ell m}(\Omega)\, d^3x.$$

  These are the *Electrostatic Multipole Moments*.

Physical Interpretation

- For $\ell = 0$ we have

$$q_{00} = \int_V \frac{1}{\sqrt{4\pi}}\,\rho(\vec{\boldsymbol{x}})\,d^3x = \frac{q}{\sqrt{4\pi}}.$$

- Note $Y_{\ell -m} = (-1)^m Y^*_{\ell m}$ implies $q_{\ell -m} = (-1)^m q^*_{\ell m}$.
- $\ell = 1$, Cartesian coordinates:

$$\vec{\boldsymbol{p}} = \int_V \vec{\boldsymbol{x}}\,\rho(\vec{\boldsymbol{x}})\,d^3x.$$

  Note: $rY_{10} = \sqrt{\frac{3}{4\pi}}z$ and $rY_{11} = -\sqrt{\frac{3}{8\pi}}(x+iy)$.
- We can then write

$$\Phi(\vec{\boldsymbol{x}}) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{r} + \frac{\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{x}}}{r^3} + \ldots\right].$$

- You will also sometimes see Cartesian moments for $\ell = 2$ (quadrupole moments).

More on Multipole Moments

- If the character size of $\rho(\vec{\boldsymbol{x}})$ is R, then we expect $q_{\ell m} \sim QR^{\ell}$, where Q is a constant that can usually be taken to be the total charge if it is non-zero. This tells us that the series for $\Phi(\vec{\boldsymbol{x}})$ converges like $(R/r)^{\ell}$, which will be very fast for $r \gg R$.

- For this reason, we normally only consider the lowest non-zero multipole.

- Example: charges $+Q$ and $-Q$ located at $(0, 0, d/2)$ and $(0, 0, -d/2)$, respectively. By representing the charge distribution with delta functions, we find $\vec{\boldsymbol{p}} = p\hat{\boldsymbol{z}}$, with $p = Qd$.

- This result can also be found by writing out the potential as the sum of two terms and expanding in a Taylor series around the origin. The leading terms have equal but opposite signs and cancel.

What does the dipole field look like?

- The math is easiest if we take $\vec{\boldsymbol{p}} = p\hat{\boldsymbol{z}}$. Then

$$\Phi(\vec{\boldsymbol{x}}) = \frac{pz}{4\pi\epsilon_0\, r^3} = \frac{p\cos\theta}{4\pi\epsilon_0\, r^2}.$$

- Then using $\vec{\boldsymbol{E}} = -\vec{\boldsymbol{\nabla}}\Phi$ we obtain

$$E_r = \frac{2p\cos\theta}{4\pi\epsilon_0\, r^3}$$
$$E_\theta = \frac{p\sin\theta}{4\pi\epsilon_0\, r^3}$$
$$E_\phi = 0.$$

- Note that $E_r(r, \theta = \pi/2) = E_\theta(r, \theta = 0) = E_\theta(r, \theta = \pi) = 0$.
- Note that the field falls off $\propto 1/r^3$.

Interaction with an external field

- The potential energy of a charge distribution $\rho(\vec{\boldsymbol{x}})$ placed an external field $\Phi(\vec{\boldsymbol{x}})$ is given by

$$W = \int_V \rho(\vec{\boldsymbol{x}})\, \Phi(\vec{\boldsymbol{x}})\, d^3x.$$

- If Φ is slowly varying, we have $\Phi(\vec{\boldsymbol{x}}) = \Phi(0) + \vec{\boldsymbol{x}} \cdot \vec{\boldsymbol{\nabla}}\Phi(0) + \ldots$ and

$$W = q\Phi(0) - \vec{\boldsymbol{p}} \cdot \vec{\boldsymbol{E}}(0) + \ldots$$

- This is useful for interaction terms in Quantum Mechanics.

- Consider dipoles $\vec{\boldsymbol{p}}_1$ and $\vec{\boldsymbol{p}}_2$ located at $\vec{\boldsymbol{x}}_1$ and $\vec{\boldsymbol{x}}_2$, with $\vec{\boldsymbol{x}}_1 - \vec{\boldsymbol{x}}_2 \propto \hat{\boldsymbol{n}}$. This results in

$$W = \frac{\vec{\boldsymbol{p}}_1 \cdot \vec{\boldsymbol{p}}_2 - 3(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{p}}_1)(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{p}}_2)}{4\pi\epsilon_0 |\vec{\boldsymbol{x}}_1 - \vec{\boldsymbol{x}}_2|^3}.$$

Polarized Material

- Suppose that $\vec{\boldsymbol{P}}(\vec{\boldsymbol{x}})$ describes the electric dipole moment per unit volume. This should be a good model for many physical situations. How can we describe this?

- Recall the result for a point dipole $[\Phi(|\vec{\boldsymbol{x}}| \to \infty) = 0]$:

$$\Phi(\vec{\boldsymbol{x}}) = \frac{\vec{\boldsymbol{p}} \cdot (\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,')}{4\pi\epsilon_0 |\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,'|^3}.$$

- By superposition, we have

$$\Phi(\vec{\boldsymbol{x}}) = \frac{1}{4\pi\epsilon_0} \int_V \left[\frac{\rho(\vec{\boldsymbol{x}}\,')}{|\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,'|} + \frac{\vec{\boldsymbol{P}}(\vec{\boldsymbol{x}}\,') \cdot (\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,')}{|\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,'|^3} \right] d^3x'.$$

- This assumes higher moments can be neglected.

Polarized Material, continued

- Using vector calculus, one can show

$$\Phi(\vec{\boldsymbol{x}}) = \frac{1}{4\pi\epsilon_0}\int_V \frac{\rho(\vec{\boldsymbol{x}}\,') - \vec{\boldsymbol{\nabla}}' \cdot \vec{\boldsymbol{P}}(\vec{\boldsymbol{x}}\,')}{|\vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}\,'|}\, d^3x'.$$

- Here, we define ρ so that it does not include the polarization.
- Note that if $\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{P}} \neq 0$, effective charge density is created.
- Next we define $\rho_{\text{pol}} = -\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{P}}$ and note

$$\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{E}} = \frac{\rho + \rho_{\text{pol}}}{\epsilon_0} \qquad \text{or} \qquad \vec{\boldsymbol{\nabla}} \cdot \left[\vec{\boldsymbol{E}} + \frac{\vec{\boldsymbol{P}}}{\epsilon_0}\right] = \frac{\rho}{\epsilon_0}.$$

- Then

$$\vec{\boldsymbol{D}} \equiv \epsilon_0 \vec{\boldsymbol{E}} + \vec{\boldsymbol{P}} \qquad \text{and} \qquad \vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{D}} = \rho.$$

Polarizable Materials

- Many materials are polarizable when a field is applied. If the field is not too strong, a linear model is accurate. We thus define

$$\vec{\boldsymbol{P}} = \epsilon_0 \chi_e \vec{\boldsymbol{E}} \qquad \text{electric susceptibility}$$
$$\kappa = 1 + \chi_e \qquad \text{dielectric constant}$$
$$\epsilon = \epsilon_0(1 + \chi_e) = \kappa\epsilon_0.$$

- Note that empty space has $\chi_e = 0$ and the limit $\chi_e \gg 1$ behaves in many respects like conducting material.
- At a boundary between two regions (1 and 2), where $\vec{\boldsymbol{P}}$ may have a discontinuity:

$$(\vec{\boldsymbol{D}}_2 - \vec{\boldsymbol{D}}_1) \cdot \hat{\boldsymbol{n}}_{21} = \sigma \qquad \text{often zero!}$$
$$(\vec{\boldsymbol{E}}_2 - \vec{\boldsymbol{E}}_1) \times \hat{\boldsymbol{n}}_{21} = 0 \qquad \Phi \text{ usually continuous}$$

where $\hat{\boldsymbol{n}}_{21}$ points from region 1 to region 2. For the second condition, continuity of Φ results if we do not have a dipole layer, which is almost always the case, see Jackson, Eq. (1.27).